9 Mark for Review

A right triangle has sides of length $3\sqrt{2}$, $5\sqrt{2}$, and $\sqrt{68}$ units. What is the area of the triangle, in square units?

(A) $8\sqrt{2}+\sqrt{68}$

(B) $30\sqrt{68}$

(C) 30

(D) 15

www.dsatuz.com

12 Mark for Review

In triangle ABC, the measure of angle B is $90°$ and $\overline{BD}$ is an altitude of the triangle. The length of $\overline{AB}$ is 24 and the length of $\overline{AC}$ is 13 greater than the length of $\overline{AB}$. What is the value of $\frac{BC}{BD}$?

A $\frac{13}{24}$

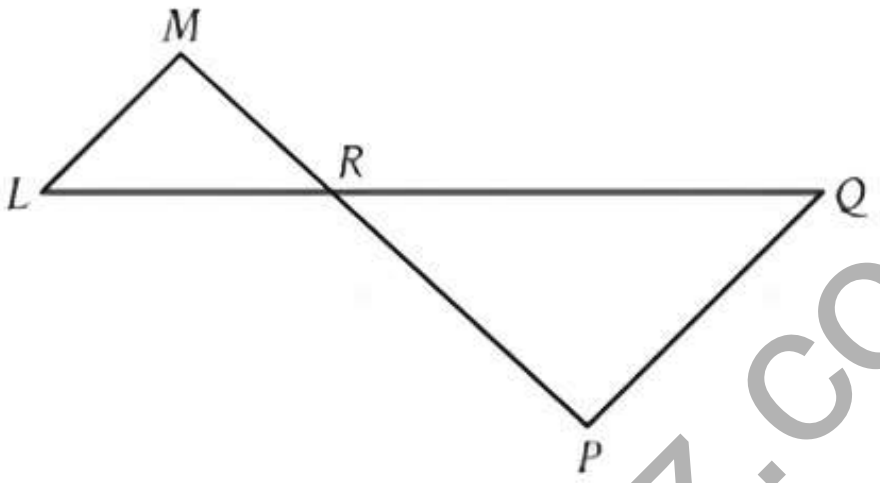
B $\frac{24}{37}$

C $\frac{37}{24}$

D $\frac{24}{13}$

www.dsatuz.com

www.dsatuz.com JOIN OUR TELEGRAM CHANNEL @DSATuz



Note: Figure not drawn to scale.

In the figure, $\overline{LQ}$ intersects $\overline{MP}$ at point R, and $\overline{LM}$ is parallel to $\overline{PQ}$. The lengths of $\overline{MR}$ and $\overline{RP}$ are 8 and 14 units, respectively. The area of $\triangle LMR$ is 36 square units. What is the area of $\triangle PQR$, in square units?

(A) $\frac{576}{49}$

(B) $\frac{144}{7}$

(C) 63

(D) $\frac{441}{4}$

www.dsatuz.com JOIN OUR TELEGRAM CHANNEL @DSATuz

20 Mark for Review

The perimeter of an equilateral triangle is 876 centimeters. The three vertices of the triangle lie on a circle. The radius of the circle is $w\sqrt{3}$ centimeters. What is the value of w?



Answer Preview:

www.dsatuz.com

19 Mark for Review

In triangle ABC, the measure of angle A is 52° and $AC = 35$. In triangle PQR, the measure of angle P is 52° and $PR = 105$. Which additional piece of information is sufficient to prove that triangle ABC is similar to triangle PQR?

(A) $AB = 30$ and $PQ = 30$.

(B) $AB = 30$ and $QR = 90$.

(C) The measures of angle B and angle R are 34° and 94°, respectively.

(D) The measures of angle B and angle Q are 52° and 34°, respectively.

www.dsatuz.com

21 Mark for Review

In triangle RST, the measure of angle R is 39°, the measure of angle S is x°, and the measure of angle T is $(5x - 3)^\circ$. Point L lies on $\overline{RS}$, point K lies on $\overline{ST}$, and $\overline{LK}$ is parallel to $\overline{RT}$. What is the measure, in degrees, of angle SKL? (Disregard the degree symbol when entering your answer.)



Answer Preview:

www.dsatuz.com

www.dsatuz.com JOIN OUR TELEGRAM CHANNEL @DSATuz

In triangle XYZ, the measure of angle X is x°, the measure of angle Y is y°, and the measure of angle Z is 31°. If $x > y > 31$, which of the following additional pieces of information is **NOT** sufficient to determine the values of x and y?

A.

The value of $31 - x - 3y$

B.

The value of $31 + 3x + y$

C.

The value of $31 - 3x + 3y$

D.

The value of $31 - 3x - 3y$

www.dsatuz.com JOIN OUR TELEGRAM CHANNEL @DSATuz

22 Mark for Review

In triangle ABC, the measure of angle B is 90° and BD is an altitude of the triangle. The length of AB is 13 and the length of AC is 19 greater than the length of AB. What is the value of $\frac{BC}{BD}$?

A $\frac{13}{32}$

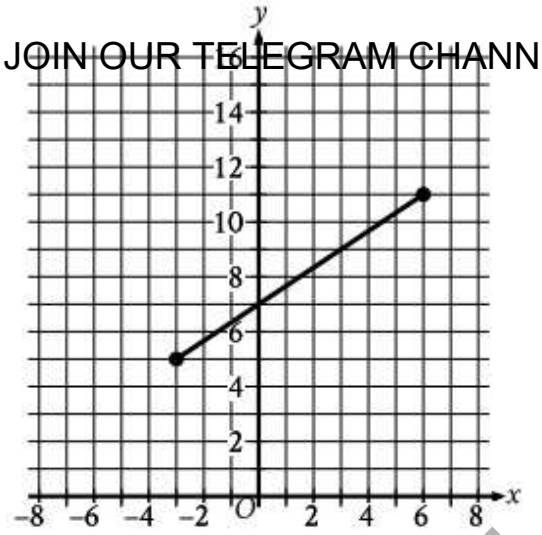
B $\frac{13}{19}$

C $\frac{19}{13}$

D $\frac{32}{13}$

www.dsatuz.com

www.dsatuz.com JOIN OUR TELEGRAM CHANNEL @DSATuz



The line segment shown in the xy-plane represents one of the legs of a right triangle. The area of this triangle is $42\sqrt{13}$ square units. What is the length, in units, of the other leg of this triangle?

A.

14

B.

28

C.

$3\sqrt{13}$

D.

$21\sqrt{13}$

www.dsatuz.com JOIN OUR TELEGRAM CHANNEL @DSATuz

www.dsatuz.com

20 Mark for Review

In isosceles triangle ABC, sides AB and AC are congruent. Point D divides side BC such that the length of $\overline{BD}$ is $\frac{2}{7}$ of the length of $\overline{BC}$. Point E lies on side AB and point F lies on side AC such that when segments DE and DF are drawn, angle BED is congruent to angle CFD. If the length of $\overline{BE}$ is 14, what is the length of $\overline{CF}$?

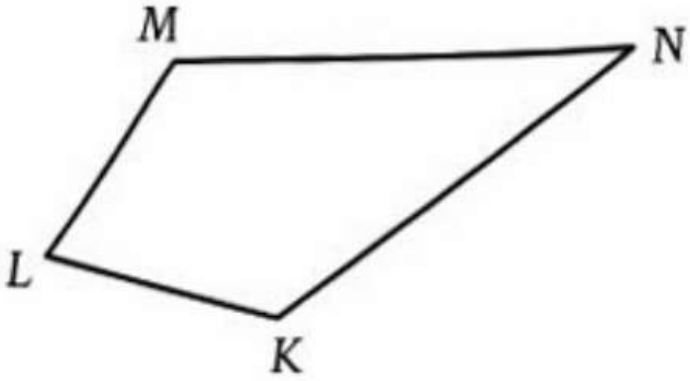
(A) 28

(B) 35

(C) 49

(D) 98

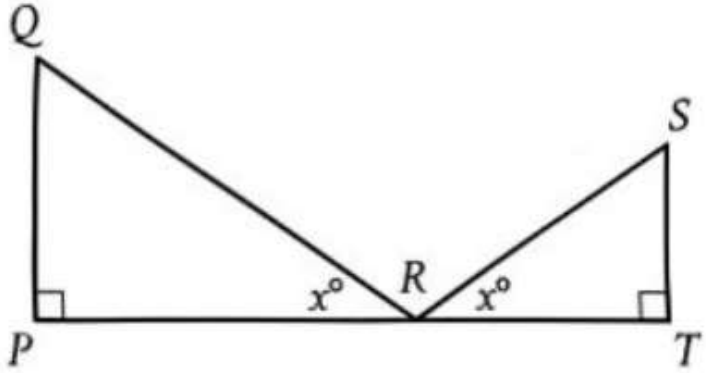
www.dsatuz.com



Note: Figure not drawn to scale.

In quadrilateral $KLMN$ shown, $KL = 3$, $LM = 3$, $KN = 27$, and $MN = 27$. Diagonals KM and LN (not shown) intersect at point G (not shown), where $GK = 1$ and $GM = 1$. If the length of diagonal LN is $\sqrt{p} + \sqrt{w}$, where p and w are integers, what is the value of $p + w$?

www.dsatuz.com



Note: Figure not drawn to scale.

$\triangle QPR$ is similar to $\triangle STR$. The lengths represented by $\overline{ST}$, $\overline{QP}$, $\overline{PR}$, and $\overline{QR}$ in the figure are 10, 16, 30, and 34, respectively. What is the length of $\overline{SR}$?

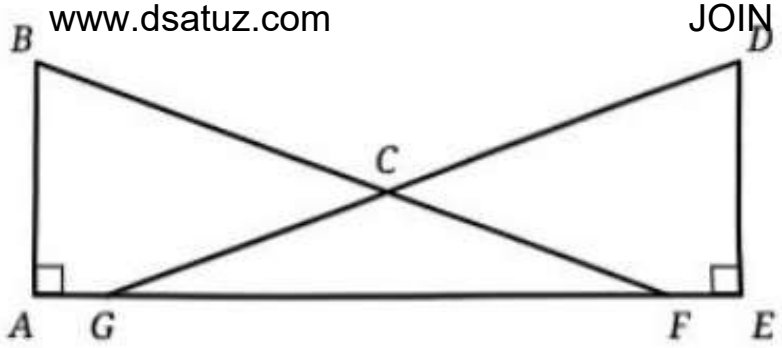
A. $\frac{340}{16}$

B. $\frac{340}{30}$

C. $\frac{160}{30}$

D. $\frac{160}{34}$

www.dsatuz.com



Note: Figure not drawn to scale.

In the figure, triangle ABF is congruent to triangle EDG, where B corresponds to D. The measure of angle DCF is 58°. What is the measure, in degrees, of angle ABF?

A. 29

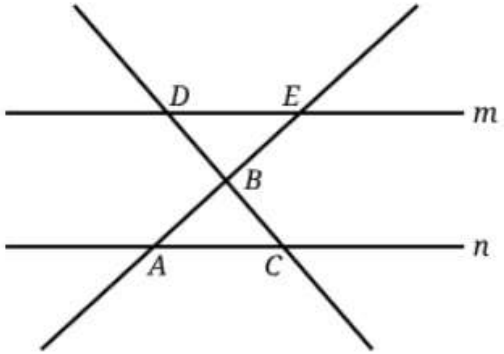
B. 32

C. 58

D. 61

www.dsatuz.com

139.



Note: Figure not drawn to scale.

In the figure, line m is parallel to line n, and lines AE and CD intersect at point B. Which additional piece of information is sufficient to prove that triangle ABC is congruent to triangle EBD?

A. $AB = 16$ and $DB = 16$.

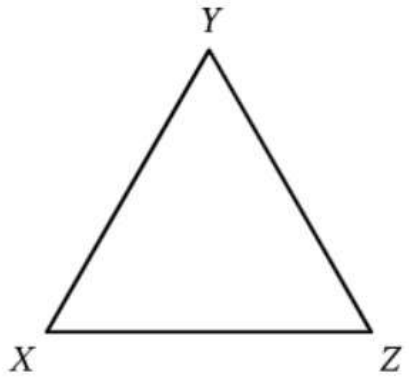
B. $AB = 16$ and $EB = 16$.

C. Triangles ABC and EBD are isosceles.

D. No additional information is necessary to determine that the two triangles are congruent.

www.dsatuz.com

114.



Note: Figure not drawn to scale.

In triangle XYZ shown, segments XY and YZ each have length 128 millimeters. Which statement is sufficient to prove that all three angles in the triangle have equal measure?

A. Segment XZ has length 64 millimeters.

B. Segment XZ has length 128 millimeters.

C. Segment XZ has length 256 millimeters.

D. Segment XZ has length 384 millimeters.

www.dsatuz.com